Supplementary Material: Exact Molecular Docking: A Machine-Checked Theory of Configuration Resolution, Complexity, and Thermodynamic Cost

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AC1	ClaimClosure.AtomicCircuitExports.AC1 paper4/DecisionQuotient/ClaimClosure.lean	AC3	ClaimClosure.AtomicCircuitExports.AC3 paper4/DecisionQuotient/ClaimClosure.lean
AC4	ClaimClosure.AtomicCircuitExports.AC4 paper4/DecisionQuotient/ClaimClosure.lean	BA1	Physics.BoundedAcquisition.BoundedRegion paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA2	Physics.BoundedAcquisition.acquisition_rate_ bound paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA3	Physics.BoundedAcquisition.acquisitions_are_ transitions paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA4	Physics.BoundedAcquisition.one_bit_per_ transition paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA5	Physics.BoundedAcquisition.resolution_reads_ sufficient paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA6	Physics.BoundedAcquisition.srank_le_ resolution_bits paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA7	Physics.BoundedAcquisition.energy_ge_srank_ cost paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA8	Physics.BoundedAcquisition.srank_one_energy_ minimum paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA10	Physics.BoundedAcquisition.counting_gap_ theorem paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
CV7	Physics.Conversation.clamp_projection_eq_iff_ same_clamped_bit paper4/DecisionQuotient/Physics/ Conversation.lean	CV8	Physics.Conversation.clampDecisionEvent_iff_ bitOps_pos paper4/DecisionQuotient/Physics/ Conversation.lean
CV9	Physics.Conversation.clamp_event_implies_ positive_energy paper4/DecisionQuotient/Physics/ Conversation.lean	DQ1	DecisionQuotient.Tractability. SampledDockingCutoff.sampled_insideCutoff_ sufficient paper4/DecisionQuotient/Tractability/ SampledDockingCutoff.lean
DQ2	DecisionQuotient.Tractability. SampledDockingGap.SampledDockingProblem.exact_ coarse_opt_agree_of_gap paper4/DecisionQuotient/Tractability/ SampledDockingGap.lean	DT22	Physics.DecisionTime.substrate_step_realizes_ decision_event paper4/DecisionQuotient/Physics/ DecisionTime.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DT23	Physics.DecisionTime.substrate_step_is_time_unit paper4/DecisionQuotient/Physics/DecisionTime.lean	DT24	Physics.DecisionTime.time_unit_law_substrate_invariant paper4/DecisionQuotient/Physics/DecisionTime.lean
EI1	ThermodynamicLift.energy_ge_kbt_nat_entropy paper4/DecisionQuotient/ThermodynamicLift.lean	IT3	DecisionQuotient.quotientEntropy_le_srank_binary paper4/DecisionQuotient/Information.lean
IT4	DecisionQuotient.numOptClasses_le_pow_srank_binary paper4/DecisionQuotient/Information.lean	L17	Leverage.compose_dof Leverage/Foundations.lean
L19	Leverage.composition_dof_additive Leverage/Theorems.lean	L43	dof_eq_srank Leverage/BridgeToDQ.lean
L44	dof_one_iff_max_leverage Leverage/Foundations.lean	L45	england_replication_inequality Leverage/BridgeToDQ.lean
L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean	L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean
L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean	L51	ssot_srank_one Leverage/BridgeToDQ.lean
L52	succ_le_two_pow Leverage/BridgeToDQ.lean	L53	sufficiency_conp_hard
L54	thermodynamic_selection Leverage/BridgeToDQ.lean	L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean
L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean	L57	Leverage.ColumnComplexityBridge.SharedCodewordCount_eq_TieBrokenRelationMoment Leverage/ColumnComplexityBridge.lean
L58	Leverage.ColumnComplexityBridge.zeroIdentityDebt_tieBrokenArgmin_of_uniform_argmin_relation_bound Leverage/ColumnComplexityBridge.lean	L59	Leverage.constrainedMolecular_energy_lower_bound Leverage/BridgeToDQ.lean
L60	Leverage.rattle_constraintObservations_card Leverage/BridgeToDQ.lean	L61	Leverage.rattle_energy_lower_bound Leverage/BridgeToDQ.lean
L62	Leverage.rattle_srank_eq_effectiveDOF Leverage/BridgeToDQ.lean	L63	Leverage.exactSufficiency_conp_core Leverage/DockingTheoryBridge.lean
L64	Leverage.exactSufficiency_hardFamily_srank_eq_n Leverage/DockingTheoryBridge.lean	L65	Leverage.molecularDocking_boundedPocket_srank_bound Leverage/DockingTheoryBridge.lean
L66	Leverage.molecularDocking_srank_bound Leverage/DockingTheoryBridge.lean	ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean
PH26	PhysicalComplexity.no_collapse_of_bounded_budget_pos_cost_exp_lb paper4/DecisionQuotient/Physics/PhysicalHardness.lean	QT1	DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean
QT2	DecisionProblem.quotientMap_preservesOpt paper4/DecisionQuotient/Quotient.lean	QT3	DecisionProblem.quotient_represents_opt_equiv paper4/DecisionQuotient/Quotient.lean
QT7	DecisionProblem.quotient_has_unique_factorization paper4/DecisionQuotient/Quotient.lean	SE1	ClaimClosure.SE1 paper4/DecisionQuotient/ClaimClosure.lean
SE2	ClaimClosure.SE2 paper4/DecisionQuotient/ClaimClosure.lean	SE3	ClaimClosure.SE3 paper4/DecisionQuotient/ClaimClosure.lean
SE4	ClaimClosure.SE4 paper4/DecisionQuotient/ClaimClosure.lean	SE5	ClaimClosure.SE5 paper4/DecisionQuotient/ClaimClosure.lean
SE6	ClaimClosure.SE6 paper4/DecisionQuotient/ClaimClosure.lean	W1	Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean
W2	Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean	W3	Physics.integrity_is_centroid paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean
W4	Physics.wasserstein_bridge paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean	WM4	Physics.WolpertMismatch.mismatchNatLowerBound_pos_of_exists_ne paper4/DecisionQuotient/Physics/WolpertMismatch.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
WM6	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_distribution_mismatch paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP2	Physics.WolpertDecomposition.landauer_floor_plus_decomposition_lower_bound paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WP6	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_either_cited_component paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP7	Physics.WolpertDecomposition.landauer_floor_plus_structural_resource_lower_bound paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WP8	Physics.WolpertDecomposition.energy_lower_bound_increases_by_structural_resource paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP9	Physics.WolpertDecomposition.physical_grounding_bundle_with_wolpert_decomposition paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WR6	Physics.WolpertResidual.discreteResidualNatLowerBound_pos_of_asymmetry_or_oneway paper4/DecisionQuotient/Physics/WolpertResidual.lean	WR7	Physics.WolpertDecomposition.stopping_time_residual_of_discrete_edge_split paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WR10	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_finite_discrete_witness paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WR11	Physics.WolpertResidual.binaryEncodedResidualNatLowerBound_eq_one paper4/DecisionQuotient/Physics/WolpertResidual.lean
WR12	Physics.WolpertDecomposition.effective_model_ge_landauer_plus_one_of_binary_encoded_residual_example paper4/DecisionQuotient/Physics/WolpertDecomposition.lean		

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 4.4: Bounded-Pocket Low-Rank Regime	L65
Corollary 5.7: Decision-Entropy Bound from Spacetime and Energy Budget	IT3, BA1, BA2, BA5, BA6, EI1, L43
Corollary 5.8: Independent Composition Budget Law	L17, L19, BA1, BA2, BA7, BA5, BA6, L43
Corollary 5.26: RATTLE Holonomic-Constraint Landauer Floor	L60, L61, L62
Corollary 5.4: Unique Minimum-Cost Regime	BA8, L54, L55
Corollary 3.3: Higher-Rank Regime	L46, L51
Corollary 3.2: Rank-One Regime	L51
Proposition 5.10: Two-Level Atomic Realization	AC1, AC3, AC4
Proposition 5.22: Repeated Binary Mismatch Work Law	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.21: Binary Mismatch Strengthens the Energy–Information Coefficient	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.20: Explicit Binary Mismatch Example	BA5, BA6, WP2, WM4, L43
Proposition 5.15: Repeated Residual-Example Work Law	IT3, BA5, BA6, WR12, WR11, L43
Proposition 5.14: Explicit Two-State Residual Example	IT3, BA5, BA6, WR12, WR11, L43
Proposition 2.5: Bounded Region	BA1

Paper claim	**Lean handle**
Proposition 5.18: Canonical Wolpert Grounding Bundle	`BA5`, `BA6`, `WP9`, `L43`
Definition 2.3: Bounded Decision System	`L17`, `L19`
Proposition 5.16: Unified Energy–Information Hierarchy	`IT3`, `BA5`, `BA6`, `WR12`, `WP2`, `WM4`, `WR11`, `L43`
Proposition 3.7: Finite Compression-Relation Bridge	`L57`, `L58`
Proposition 5.13: Finite Discrete Residual Witness	`WR10`, `WR7`, `WR6`
Proposition 5.23: Positive Heat and Bounded Lifetime	`SE1`, `SE2`, `SE3`, `SE4`, `SE5`
Proposition 5.24: Finite Lifetime Throughput Bound	`SE5`, `IT3`, `L43`
Definition 2.13: Canonical Decision Problem	`QT2`, `QT7`, `QT1`, `QT3`
Proposition 5.25: Speed-Heat Tradeoff	`SE6`
Proposition 5.19: Strict Canonical Energy Above the Landauer Floor	`BA5`, `BA6`, `WP6`, `L43`
Proposition 5.12: Theorem-Level Strict Overhead Branches	`BA5`, `BA6`, `WM6`, `WP6`, `WR10`, `L43`
Proposition 5.17: Structural-Resource Overhead	`BA5`, `BA6`, `WP8`, `WP7`, `L43`
Proposition 5.11: Substrate Step is Unit Interface Time	`DT23`, `DT22`, `DT24`
Proposition 5.9: Threshold Channel Realization	`CV8`, `CV9`, `CV7`
Theorem 2.6: Bounded Acquisition Rate	`BA1`, `BA2`
Theorem 5.6: Decision-Class Bound from Spacetime and Energy Budget	`IT4`, `IT3`, `BA1`, `BA2`, `BA5`, `BA6`, `EI1`, `L43`
Theorem 6.1: Coherent Single-Source Regime	`ORA1`
Theorem 2.4: Counting Gap	`BA10`
Theorem 2.7: Discrete Acquisition	`BA3`
Theorem 3.1: DOF–Structural-Rank Identity	`L43`
Theorem 5.3: Energy–Information Duality	`IT3`, `EI1`, `L43`
Theorem 5.1: Rank Controls Exact-Resolution Cost	`BA7`, `BA6`, `L43`
Theorem 6.8: Finite Replication Entropy Gap	`L45`
Theorem 3.6: Decision-Entropy Bound	`IT3`, `L43`
Theorem 4.1: General Hardness Core for Exact Sufficiency	`L63`
Theorem 6.9: Finite-Budget No-Collapse	`PH26`
Theorem 6.7: Convergence	`L43`, `L44`, `L47`, `L55`
Theorem 4.2: Maximal Structural Rank in the Hard Family	`L64`
Theorem 3.4: Minimum Physical Bit Operations	`BA5`, `BA6`, `L43`, `L46`
Theorem 4.3: Cutoff-Local Structural-Rank Bound for Exact Docking	`L66`
Theorem 3.5: Decision-Class Bound	`IT4`, `L43`
Theorem 2.8: One Transition, One Bit	`BA4`
Theorem 6.3: Rank Identification	`L43`, `L46`, `L51`
Theorem 5.2: Rank-One Ground State	`BA8`, `L54`, `L55`
Theorem 2.9: Resolution Requires a Sufficient Coordinate Set	`BA5`
Theorem 4.5: Sampled Exact-Coarse Winner Preservation	`DQ2`

Paper claim	Lean handle
Theorem 4.6: Inside-Cutoff Sufficiency for Sampled Docking	DQ1
Theorem 6.5: Thermodynamic Selection	BA8, L49, L54, L55
Theorem 5.5: Exact-Resolution Time Lower Bound	BA1, BA2, BA5, BA6, L43
Theorem 6.4: Tractable Sufficiency at Rank One	L47, L53, L56

Auto summary: mapped 54/54 (full=54, derived=0, unmapped=0).

3 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor: bounded-pocket-regime	unspecified	-	L65
cor: budget-entropy-bound	unspecified	-	IT3, BA1, BA2, BA5, BA6, EI1, L43
cor: composition-budget-law	unspecified	-	L17, L19, BA1, BA2, BA7, BA5, BA6, L43
cor: holonomic-landauer-floor	unspecified	-	L60, L61, L62
cor: minimum-cost-regime	unspecified	-	BA8, L54, L55
cor:rank-above-one	unspecified	-	L46, L51
cor:rank-one	unspecified	-	L51
prop: atomic-realization	unspecified	-	AC1, AC3, AC4
prop: binary-mismatch-cumulative-work	unspecified	-	IT3, BA5, BA6, WP2, WM4, L43
prop: binary-mismatch-energy-information	unspecified	-	IT3, BA5, BA6, WP2, WM4, L43
prop: binary-mismatch-example	unspecified	-	BA5, BA6, WP2, WM4, L43
prop: binary-residual-cumulative-work	unspecified	-	IT3, BA5, BA6, WR12, WR11, L43
prop: binary-residual-example	unspecified	-	IT3, BA5, BA6, WR12, WR11, L43
prop:bounded-region	unspecified	-	BA1
prop: canonical-wolpert-bundle	unspecified	-	BA5, BA6, WP9, L43
prop:dof-additive	unspecified	-	L17, L19
prop:ei-hierarchy	unspecified	-	IT3, BA5, BA6, WR12, WP2, WM4, WR11, L43
prop: finite-compression-bridge	unspecified	-	L57, L58
prop: finite-discrete-residual	unspecified	-	WR10, WR7, WR6
prop:finite-lifetime	unspecified	-	SE1, SE2, SE3, SE4, SE5
prop: lifetime-throughput	unspecified	-	SE5, IT3, L43
prop: optimizer-quotient	unspecified	-	QT2, QT7, QT1, QT3
prop: speed-heat-tradeoff	unspecified	-	SE6
prop: strict-canonical-energy	unspecified	-	BA5, BA6, WP6, L43
prop:strict-overhead	unspecified	-	BA5, BA6, WM6, WP6, WR10, L43
prop: structural-resource-overhead	unspecified	-	BA5, BA6, WP8, WP7, L43
prop: substrate-time-law	unspecified	-	DT23, DT22, DT24

prop: threshold-channel	unspecified	-	CV8, CV9, CV7
thm: bounded-acquisition	unspecified	-	BA1, BA2
thm: budget-class-bound	unspecified	-	IT4, IT3, BA1, BA2, BA5, BA6, EI1, L43
thm: coherent-single-source	unspecified	-	ORA1
thm:counting-gap	unspecified	-	BA10
thm: discrete-acquisition	unspecified	-	BA3
thm:dof-srank	unspecified	-	L43
thm:energy-entropy	unspecified	-	IT3, EI1, L43
thm:energy-rank	unspecified	-	BA7, BA6, L43
thm:england	unspecified	-	L45
thm:entropy-bound	unspecified	-	IT3, L43
thm: exact-sufficiency-hardness-core	unspecified	-	L63
thm: finite-budget-no-collapse	unspecified	-	PH26
thm:five-way	unspecified	-	L43, L44, L47, L55
thm: hard-family-srank	unspecified	-	L64
thm: min-bit-operations	unspecified	-	BA5, BA6, L43, L46
thm: molecular-docking-srank-bound	unspecified	-	L66
thm:numopt-bound	unspecified	-	IT4, L43
thm: one-transition-one-bit	unspecified	-	BA4
thm: rank-identification	unspecified	-	L43, L46, L51
thm:rank-one-ground	unspecified	-	BA8, L54, L55
thm: resolution-sufficient	unspecified	-	BA5
thm: sampled-docking-gap	unspecified	-	DQ2
thm: sampled-inside-cutoff-sufficient	unspecified	-	DQ1
thm: thermodynamic-selection	unspecified	-	BA8, L49, L54, L55
thm:time-lower-bound	unspecified	-	BA1, BA2, BA5, BA6, L43
thm: tractable-rank-one	unspecified	-	L47, L53, L56

Auto summary: indexed 54 claims by hardness profile (unspecified=54).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- `Leverage/BridgeToDQ.lean`
- `Leverage/FiveWayEquivalence.lean`
- `Leverage/Physical.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)